

I agree, this value should be much larger in order to tell a story that treatment helps to control the ed rate.

GENROCKET Dataset Summary

| Metric                                      | Current value |
|---------------------------------------------|---------------|
| Total members                               | 1,000         |
| Treated members                             | 325           |
| Untreated/control members                   | 675           |
| Treatment rate                              | 32.5%         |
| ED outcome events                           | 98            |
| Outcome prevalence                          | 9.8%          |
| Treated observed ED rate                    | 21.5%         |
| Control observed ED rate                    | 4.1%          |
| Final predictors before one-hot encoding    | 56            |
| Continuous/count numeric predictors         | 18            |
| Binary indicator predictors                 | 22            |
| Multi-level categorical predictors          | 16            |
| Model matrix columns after one-hot encoding | 1,245         |

GENROCKET added variables notes:

- Variables I am comfortable with GENROCKET adding:
  - id
  - member\_id
  - intervention\_flag
- Notes on accepted variables:
  - id and member\_id are acceptable row/member identifiers.
  - intervention\_flag is acceptable and is corresponding properly with intervention\_type.
  - outcome\_ed\_90d is acceptable as the modeled outcome field, but it should continue to be checked against the expected outcome and dependency logic.

**Commented [ND1]:** A little low, we want this closer to 40%

**Commented [RH2]:** What is the ED rate before intervention for this group? Check these two variables  
ed\_visits\_last\_30d  
ed\_visits\_last\_6m

**Commented [ND3R2]:** For this treated group before intervention:

ed\_visits\_last\_30d:  
mean: 0.26  
median: 0

ed\_visits\_last\_6m:  
mean: 2.93  
median: 3

**Commented [ND4R2]:** Comparing this to the baseline utilization for the untreated group:

ed\_visits\_last\_30d:  
mean: 0.05  
median: 0

ed\_visits\_last\_6m:  
mean: 2.50  
median: 3

**Commented [ND5R2]:** Ask Kristen if she can provide insight on this.

**Commented [RH6]:** 4.1% looks very low for a case-management-eligible population

**Commented [ND7R6]:** Agreed. For the synthetic data adjustment, I think the untreated/control group should have a higher probability of outcome\_ed\_90d = 1, especially among members with prior ED utilization. A 4.1% control outcome rate seems low for this population and may cause the model to underestimate untreated counterfactual ED risk. Rather than increasing outcomes uniformly, I would suggest recalibrating the untreated ED outcome probability based on baseline risk factors such as ed\_visits\_last\_30d and ed\_visits\_last\_6m, so higher-utilization untreated members have a more realistic chance of an ED outcome.

- Variables that need more clarification:
  - state
  - Propensity\_Score
  - true\_treatment\_effect
- Notes on variables needing clarification:
  - state should be added to the specification if it is intended for modeling or reporting; otherwise, consider removing it from the dataset.
  - Propensity\_Score appears to be a modeling or benchmark field. Its purpose should be documented, and it should be excluded from model features unless it is only being used for validation.
  - true\_treatment\_effect appears to be a modeling or benchmark field. Its purpose should be documented, and it should be excluded from model features unless it is only being used for validation.

**Commented [RH8]:** At state level, the distribution of many variables won't be uniform- many states may have very few observations for certain variables. Suggestion: reduce it to a smaller number of meaningful service regions. If we were to keep all 50 states, we need to explicitly encode state-level differences. Thoughts?

**Commented [ND9R8]:** I agree. Keeping state, county, and service region may be too granular and could create sparsity issues. I suggest we simplify the geography variables by assuming all members are from the same state, removing county, and keeping service region as the main geography variable. Service region should still capture meaningful operational differences for case management, while reducing dimensionality and avoiding very small county-level buckets.

**Commented [RH10]:** Nicky, could you run a summary on your side to double check my results? 1. The average propensity score is **0.31**. **0.46** for intervention group, **0.24** for untreated members average. 2. The propensity score distribution is much more concentrated at the low end. Currently, **50% of members have propensity scores  $\leq 0.20$** , whereas our target median was approximately **0.42**. suggestion: shift the distribution upward so that more members fall within the **0.25–0.75** overlap region. Thought? 3. Many members share identical propensity scores (e.g., **0.15, 0.18, 0.20, 0.48, 0.75**). We would expect a much smoother distribution of values like continuous probabilities. 4. Let **Genrocket confirm the generation order**. The synthetic data should be generated in the following sequence: **Baseline Variables  $\rightarrow$  Propensity Score  $\rightarrow$  Probabilistic Treatment Assignment** rather than assigning treatment first and then generating the propensity score. This ordering reflects how observational healthcare data are generated and aligns with the assumptions of DR Learners, T-/X-Learners, and Causal Forests.

**Commented [ND11R10]:**  
I ran a summary on my side to verify the propensity score results.

The average propensity score is 0.31 overall, 0.46 for treated/intervention members, and 0.24 for untreated members. The median propensity score is 0.20. Approximately 58.7% of members have propensity scores  $\leq 0.20$ , while 26.0% fall within the 0.25-0.75 overlap region.

I also see repeated propensity score values. There are 21 unique propensity score values across 1000 members, meaning only 2.1% of rows have unique propensity scores. This supports the concern that the propensity scores may have been generated in discrete buckets rather than as smoother continuous probabilities.

Based on this, I agree with the recommendation to shift the propensity score distribution upward, increase the ...

### Distribution Issues

| Variable       | Expected Distribution                             | GENROCKET Result                                     | Status | Finding                                                                                                             |
|----------------|---------------------------------------------------|------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------|
| admits_last_6m | 0: 85%<br>1: 10%<br>2: 3%<br>3+: 2%               | 0: 30.3%; 1: 15.5%; 2: 18.3%; 3+: 35.9%              | Fail   | Bucket distribution differs from expected; max absolute bucket difference is 49.7% and average difference is 24.9%. |
| age            | 18-34: 20%<br>35-49: 25%<br>50-64: 35%<br>65+:20% | 18-34: 27.5%; 35-49: 21.7%; 50-64: 20.8%; 65+: 30.0% | Fail   | Bucket distribution differs from expected; max absolute bucket difference is 14.2% and average difference is 8.8%.  |

|                             |                                                                               |                                                                                                                                                                                                            |      |                                                                                                                       |
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| anxiety_flag                | 18%-28% propensity 1                                                          | 12.6%<br><br><b>This is about right for Acentra clients (8 – 15%)</b>                                                                                                                                      | Fail | Observed 12.6% is outside expected 18.0% to 28.0%.                                                                    |
| asthma_flag                 | 12%-18% propensity 1                                                          | 7.6%<br><br><b>This is about right for Acentra clients (4 – 12%)</b>                                                                                                                                       | Fail | Observed 7.6% is outside expected 12.0% to 18.0%.                                                                     |
| behavioral_health_risk_flag | 20-30% propensity 1<br><br><b>20-25% is more in line with Acentra clients</b> | 41.4%                                                                                                                                                                                                      | Fail | Observed 41.4% is outside expected 20.0% to 30.0%.                                                                    |
| case_manager_name           | CM_001: 24%<br>CM_002: 24%<br>CM_003: 23%<br>CM_004: 24%<br>Unassigned: 5%    | LEEANN GAUDETTE: 0.1%;<br>MISTI GOSNELL: 0.1%;<br>BRITTANY LUSTER: 0.1%;<br>GOLDIE PRENTICE: 0.1%;<br>TWILA MONTEMAYOR: 0.1%;<br>KATHLEEN SORENSON: 0.1%;<br>HOLLY TRUMP: 0.1%;<br>JACQUELYNE MORGAN: 0.1% | Fail | Distribution shape differs from expected; max absolute top-share difference is 23.9% and average difference is 19.9%. |
| chf_flag                    | 14% propensity 1<br><br><b>8-15% is more in line with Acentra clients</b>     | 4.4%                                                                                                                                                                                                       | Fail | Observed 4.4% is outside expected 11.5% to 16.5%.                                                                     |
| ckd_flag                    | 15% propensity 1<br><br><b>9-15% is more in line with Acentra clients</b>     | 8.5%                                                                                                                                                                                                       | Fail | Observed 8.5% is outside expected 12.5% to 17.5%.                                                                     |
| client_contract             | Medicaid MCO: 40%                                                             | Contract1: 100.0%                                                                                                                                                                                          | Fail | Distribution shape differs from expected; max                                                                         |

|                 |                                                                                                                                                                                                                                                          |                                                                                                                                                      |      |                                                                                                                      |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------------------------------------------------------------------------------------------------------------------|
|                 | Medicaid Expansion: 18%<br>Dual Eligibility: 22%<br>MLTSS: 8%<br>Behavioral Health: 6%<br>Health Home: 3%<br>Maternal Health: 2%<br>SUD: 1%<br>FFS/PCCM: 0-3% optional                                                                                   |                                                                                                                                                      |      | absolute top-share difference is 60.0% and average difference is 13.5%.                                              |
| copd_flag       | 11% propensity 1                                                                                                                                                                                                                                         | 10.4%                                                                                                                                                | Pass |                                                                                                                      |
| county          | County_A_Large_Urban: 28%<br>County_B_Large_urban: 18%<br>County_C_Suburban: 14%<br>County_D_Suburban: 10%<br>County_E_Small_Metro: 8%<br>County_F_Rural: 6%<br>County_G_Rural: 5%<br>County_H_Rural: 4%<br>County_I_Frontier: 2%<br>Unknown/Missing: 5% | Marion: 4.5%;<br>Franklin: 4.3%;<br>Carson City: 3.0%;<br>Thurston: 2.5%;<br>Santa Fe: 2.4%;<br>Juneau: 2.4%;<br>Hartford: 2.4%;<br>Montgomery: 2.4% | Fail | Distribution shape differs from expected; max absolute top-share difference is 23.5% and average difference is 7.2%. |
| depression_flag | 20%-30% propensity 1<br><b>20-30% is in line with Acentra clients</b>                                                                                                                                                                                    | 11.6%                                                                                                                                                | Fail | Observed 11.6% is outside expected 20.0% to 30.0%.                                                                   |

|                          |                                                                                                                                                                                                                                      |                                            |       |                                                                                                                      |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-------|----------------------------------------------------------------------------------------------------------------------|
| diabetes_flag            | 28% propensity 1<br><br><b>25 – 30% is in line with Acentra clients</b>                                                                                                                                                              | 20.2%                                      | Fail  | Observed 20.2% is outside expected 25.5% to 30.5%.                                                                   |
| dual_eligible            | 18%-25% propensity 1                                                                                                                                                                                                                 | 25.0%                                      | Pass  |                                                                                                                      |
| ed_visits_last_30d       | 0: 92%<br><br>1: 6%<br><br>2: 1.5%<br><br>3+: 0.5%                                                                                                                                                                                   | 0: 90.2%; 1: 8.2%;<br>2: 1.6%; 3+: 0.0%    | Pass  |                                                                                                                      |
| ed_visits_last_6m        | 0: 70%<br><br>1: 15%<br><br>2: 8%<br><br>3+: 7%<br><br><b>Use this</b>                                                                                                                                                               | 0: 15.8%; 1: 14.5%; 2: 15.5%;<br>3+: 54.2% | Fail  | Bucket distribution differs from expected; max absolute bucket difference is 54.2% and average difference is 27.3%.  |
| high_cost_drug_flag      | 5%-10% propensity 1                                                                                                                                                                                                                  | 4.7%                                       | Watch | Observed 4.7% is outside expected 5.0% to 10.0%.                                                                     |
| housing_instability_flag | 8%-15% propensity 1                                                                                                                                                                                                                  | 10.0%                                      | Pass  |                                                                                                                      |
| language                 | English: 78%<br><br>Spanish: 13%<br><br>Arabic: 1.5%<br><br>Chinese: 1.5%<br><br>Vietnamese: 1%<br><br>Haitian Creole: 0.8%<br><br>Russian: 0.7%<br><br>Tagalog: 0.7%<br><br>ASL: 0.3%<br><br>Other: 1%<br><br>Unknown/Missing: 0.5% | ENG: 75.0%;<br>OTHER: 25.0%                | Watch | Distribution shape differs from expected; max absolute top-share difference is 12.0% and average difference is 2.1%. |

|                           |                                                                                                              |                                            |       |                                                                                                                       |
|---------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------|
| living_alone_flag         | 0: 74%<br>1: 23%<br>Unknown/Missing: 3%                                                                      | N: 68.8%; Y: 31.2%                         | Watch | Distribution shape differs from expected; max absolute top-share difference is 8.2% and average difference is 5.5%.   |
| observation_stays_last_6m | 0: 91%<br>1: 6%<br>2: 2%<br>3+: 1%<br><br><b>Use this</b>                                                    | 0: 30.3%; 1: 15.5%; 2: 18.3%; 3+: 35.9%    | Fail  | Bucket distribution differs from expected; max absolute bucket difference is 60.7% and average difference is 30.3%.   |
| opioid_flag               | 5%-10% propensity 1<br><br><b>Use this</b>                                                                   | 14.9%                                      | Fail  | Observed 14.9% is outside expected 5.0% to 10.0%.                                                                     |
| pcp_visits_last_6m        | 0: 15%<br>1-2: 30%<br>3-5: 35%<br>6+: 20%<br><br><b>Use this</b>                                             | 0: 22.2%; 1-2: 52.8%; 3-5: 20.8%; 6+: 4.2% | Fail  | Bucket distribution differs from expected; max absolute bucket difference is 22.8% and average difference is 15.0%.   |
| polypharmacy_flag         | 15%-25% propensity 1                                                                                         | 17.6%                                      | Pass  |                                                                                                                       |
| pregnancy_flag            | 3%-6% propensity 1<br><br><b>Use this</b>                                                                    | 21.0%                                      | Fail  | Observed 21.0% is outside expected 3.0% to 6.0%.                                                                      |
| program                   | Complex Care Management: 30%<br><br>High_Risk Case Management: 20%<br><br>Behavioral Health Integration: 14% | Program1: 100.0%                           | Fail  | Distribution shape differs from expected; max absolute top-share difference is 70.0% and average difference is 14.0%. |

|                           |                                                                                                                                                                               |                                                                                    |      |                                                                                                                       |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------|-----------------------------------------------------------------------------------------------------------------------|
|                           | Health Home: 10%<br><br>LTSS: 8%<br><br>Dual Eligible Care Management: 8%<br><br>Transitions of Care: 4%<br><br>ED Diversion: 3%<br><br>Maternal/Pregnancy: 2%<br><br>SUD: 1% |                                                                                    |      |                                                                                                                       |
| rx_count_last_6m          | 0: 10%<br><br>1-2: 12%<br><br>3-5: 18%<br><br>6-11: 25%<br><br>12-23: 22%<br><br>24-35: 8%<br><br>36+: 5%<br><br><b>Use this</b>                                              | 0: 43.7%; 1-2: 16.9%; 3-5: 26.0%; 6-11: 10.6%; 12-23: 2.8%; 24-35: 0.0%; 36+: 0.0% | Fail | Bucket distribution differs from expected; max absolute bucket difference is 33.7% and average difference is 13.3%.   |
| service_region            | Urban Core: 45%<br><br>Suburban: 25%<br><br>Rural: 20%<br><br>Frontier: 5%<br><br>Unknown/Missing: 5%                                                                         | Region1: 100.0%                                                                    | Fail | Distribution shape differs from expected; max absolute top-share difference is 55.0% and average difference is 22.0%. |
| specialist_visits_last_6m | 0: 30%<br><br>1-2: 30%<br><br>3-5: 25%<br><br>6+: 15%<br><br><b>Use this</b>                                                                                                  | 0: 67.0%; 1-2: 22.4%; 3-5: 8.8%; 6+: 1.8%                                          | Fail | Bucket distribution differs from expected; max absolute bucket difference is 37.0% and average difference is 18.5%.   |

|                             |                                                                                                                                                                       |                                                                                                                                                                |                |                                                                                                                     |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------|
| substance_use_flag          | 8%-15% propensity 1<br><br><b>Use this</b>                                                                                                                            | 29.2%                                                                                                                                                          | Fail           | Observed 29.2% is outside expected 8.0% to 15.0%.                                                                   |
| total_cost_last_6m          | 0\$-499\$: 10%<br>500\$-1999\$: 20%<br>\$2000-4999\$: 25%<br>\$5000-9999\$: 20%<br>10000\$-24999\$: 15%<br>25000\$-49999\$: 7%<br>50000\$+: 3%<br><br><b>Use this</b> | 0\$-499\$: 94.3%;<br>500\$-1999\$: 5.6%;<br>\$2000-4999\$: 0.0%;<br>\$5000-9999\$: 0.0%;<br>10000\$-24999\$: 0.0%;<br>25000\$-49999\$: 0.0%;<br>50000\$+: 0.0% | Fail           | Bucket distribution differs from expected; max absolute bucket difference is 84.3% and average difference is 24.1%. |
| transportation_barrier_flag | 12-20% propensity 1                                                                                                                                                   | 14.6%                                                                                                                                                          | Pass           |                                                                                                                     |
| current_risk_score          | 0-19: 8%<br>20-39: 22%<br>40-59: 35%<br>60-79: 25%<br>80-100: 10%                                                                                                     |                                                                                                                                                                | Missing column | The target variable is absent from GENROCKET.csv.                                                                   |
| percolator_clinical_score   | 0-19: 12%<br>20-39: 28%<br>40-59: 30%<br>60-79: 20%<br>80-100: 10%                                                                                                    | 0-19: 100.0%; 20-39: 0.0%; 40-59: 0.0%; 60-79: 0.0%; 80-100: 0.0%                                                                                              | Fail           | Bucket distribution differs from expected; max absolute bucket difference is 88.0% and average difference is 35.2%. |
| percolator_sdoh_score       | 0-19: 35%<br>20-39: 25%<br>40-59: 20%<br>60-79: 12%                                                                                                                   | 0-19: 100.0%; 20-39: 0.0%; 40-59: 0.0%; 60-79: 0.0%; 80-100: 0.0%                                                                                              | Fail           | Bucket distribution differs from expected; max absolute bucket difference is 65.0% and average                      |



|                              |                                                                                                                     |                                                                                                                                      |       |                                                                                                                      |
|------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------------------------------------------------------------------------------------|
|                              | 80-100: 8%                                                                                                          |                                                                                                                                      |       | difference is 26.0%.                                                                                                 |
| percolator_utilization_score | 0-19: 35%<br>20-39: 30%<br>40-59: 18%<br>60-79: 10%<br>80-100: 7%                                                   | 0-19: 100.0%; 20-39: 0.0%; 40-59: 0.0%; 60-79: 0.0%; 80-100: 0.0%                                                                    | Fail  | Bucket distribution differs from expected; max absolute bucket difference is 65.0% and average difference is 26.0%.  |
| food_insecurity_flag         | 15%-25% propensity 1                                                                                                | 20.0%                                                                                                                                | Pass  |                                                                                                                      |
| gender                       | Female: 56%<br>Male: 43%<br>Missing/Not reported: 1%                                                                | M: 49.0%; F: 49.0%; U: 1.4%; O: 0.6%                                                                                                 | Watch | Distribution shape differs from expected; max absolute top-share difference is 7.0% and average difference is 4.5%.  |
| med_adherence_pdc            | <0.5: 10%<br>0.5-0.79: 30%<br>>=0.80: 60%<br><br><b>Use This</b>                                                    | <0.5: 0.3%; 0.5-0.79: 14.0%;<br>>=0.80: 85.7%                                                                                        | Fail  | Bucket distribution differs from expected; max absolute bucket difference is 25.7% and average difference is 17.1%.  |
| plan_type                    | Comprehensive_MCO: 70%<br>MLTSS: 8%<br>D-SNP: 7%<br>PIHP: 5%<br>PAHP: 3%<br>PCCM: 3%<br>PCCM Entity: 1%<br>PACE: 1% | Medicaid: 30.0%;<br>Medicare: 30.0%;<br>Medicare Advantage: 20.0%;<br>Commercial: 15.0%;<br>Marketplace: 5.0%<br><br><b>Use this</b> | Fail  | Distribution shape differs from expected; max absolute top-share difference is 40.0% and average difference is 9.4%. |

|                           |                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                      |       |                                                                                                                       |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------|
|                           | Limited Benefit Plan: 1%<br>FFS: 1%                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                      |       |                                                                                                                       |
| utilities_insecurity_flag | 12% propensity 1                                                                                                                                                                                                                                                                                                                                                      | 14.5%                                                                                                                                | Pass  |                                                                                                                       |
| risk_tier                 | Low: 25%<br>Moderate: 35%<br>High: 25%<br>Very High: 10%<br>Rising Risk: 3%<br>Impactable High Risk: 2%                                                                                                                                                                                                                                                               | Low: 67.0%;<br>Medium: 22.4%;<br>High: 8.8%; Very High: 1.8%<br><br><b>This is right for Acentra clients</b>                         | Fail  | Distribution shape differs from expected; max absolute top-share difference is 32.0% and average difference is 10.7%. |
| intervention_type         | None / Not Enrolled: 60.0% Comprehensive Care Management: 12.0% Transitions of Care / Post-Discharge: 7.0% SDOH / Community Resource Navigation: 6.0% Behavioral Health / SUD Coordination: 5.0% Pharmacy / Medication Management: 4.0% LTSS / Home-Based Support Coordination: 3.0% Maternal / High-Risk Pregnancy: 1.5% ED Diversion / Urgent Care Navigation: 1.5% | nan: 67.5%; Care Management: 9.9%; Disease Management: 8.3%; Social Services: 5.9%; Pharmacy Review: 4.8%; Behavioral Outreach: 3.6% | Watch | Distribution shape differs from expected; max absolute top-share difference is 7.5% and average difference is 2.0%.   |
| intervention_days_active  | 0 / No Active Intervention: 60.0% 1-14 days: 4.0% 15-30 days: 6.0% 31-90 days: 10.0% 91-180 days: 14.0% 181-                                                                                                                                                                                                                                                          | 0 / No Active Intervention: 0.0%; 1-14 days: 0.0%; 15-30 days: 1.8%; 31-90 days: 76.7%; 91-180                                       | Fail  | Bucket distribution differs from expected; max absolute bucket difference is 66.7% and average                        |

|                       |                                                                                                                                                   |                                                                                                                                                    |      |                                                                                                                     |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------|
|                       | 365 days: 5.0% 366+ days: 1.0%                                                                                                                    | days: 15.7%; 181-365 days: 5.8%; 366+ days: 0.0%                                                                                                   |      | difference is 19.8%.                                                                                                |
| touches_per_month     | 0 / Not Selected for Outreach: 50.0% 1 attempt: 7.0% 2 attempts: 10.0% 3 attempts: 12.0% 4-5 attempts: 14.0% 6+ attempts: 7.0%                    | 0 / Not Selected for Outreach: 33.0%; 1 attempt: 34.0%; 2 attempts: 10.7%; 3 attempts: 15.2%; 4-5 attempts: 5.7%; 6+ attempts: 0.6%                | Fail | Bucket distribution differs from expected; max absolute bucket difference is 27.0% and average difference is 10.4%. |
| outreach_attempts     | 0 / Not Selected for Outreach: 50.0% 1 attempt: 7.0% 2 attempts: 10.0% 3 attempts: 12.0% 4-5 attempts: 14.0% 6+ attempts: 7.0%                    | 0 / Not Selected for Outreach: 33.0%; 1 attempt: 33.0%; 2 attempts: 1.0%; 3 attempts: 0.0%; 4-5 attempts: 5.0%; 6+ attempts: 11.7%                 | Fail | Bucket distribution differs from expected; max absolute bucket difference is 26.0% and average difference is 13.0%. |
| successful_contacts   | 0 successful contacts: 60.0% 1 successful contact: 15.0% 2 successful contacts: 10.0% 3-4 successful contacts: 10.0% 5+ successful contacts: 5.0% | 0 successful contacts: 66.2%; 1 successful contact: 2.8%; 2 successful contacts: 1.2%; 3-4 successful contacts: 9.7%; 5+ successful contacts: 3.4% | Fail | Bucket distribution differs from expected; max absolute bucket difference is 12.2% and average difference is 5.8%.  |
| avg_call_duration_min | <5 min: 5.0% 5-10 min: 10.0% 11-20 min: 15.0% 21-30 min: 7.0% >30 min: 3.0%                                                                       | <5 min: 1.0%; 5-10 min: 5.1%; 11-20 min: 16.8%; 21-30 min: 7.3%; >30 min: 2.3%                                                                     | Pass |                                                                                                                     |

|                         |                                                                                                                                                            |                                                                                                                                             |       |                                                                                                                       |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------|
| max_call_duration_min   | NULL / No Completed Phone Contact: 60.0% <10 min: 4.0% 10–20 min: 11.0% 21–30 min: 13.0% 31–45 min: 8.0% 46–60 min: 3.0% >60 min: 1.0%                     | NULL / No Completed Phone Contact: 67.5%; <10 min: 4.5%; 10-20 min: 18.4%; 21-30 min: 7.3%; 31-45 min: 2.3%; 46-60 min: 0.0%; >60 min: 0.0% | Watch | Bucket distribution differs from expected; max absolute bucket difference is 7.5% and average difference is 4.4%.     |
| notes_escalation_flag   | 0 / No: 88.0% 1 / Yes: 12.0%                                                                                                                               | N: 91.8%; Y: 8.2%                                                                                                                           | Pass  |                                                                                                                       |
| community_referral_flag | 0 / No: 75.0% 1 / Yes: 25.0%                                                                                                                               | N: 88.0%; Y: 12.0%                                                                                                                          | Fail  | Distribution shape differs from expected; max absolute top-share difference is 13.0% and average difference is 13.0%. |
| pharmacy_review_flag    | 0 / No: 80.0% 1 / Yes: 20.0%                                                                                                                               | N: 94.8%; Y: 5.2%                                                                                                                           | Fail  | Distribution shape differs from expected; max absolute top-share difference is 14.8% and average difference is 14.8%. |
| engagement_level        | Not Targeted / No Outreach: 50.0% Unable to Reach: 8.0% Declined / Opted Out: 2.0% Low Engagement: 15.0% Moderate Engagement: 15.0% High Engagement: 10.0% | High: 72.5%; Medium: 19.3%; Low: 8.2%                                                                                                       | Fail  | Distribution shape differs from expected; max absolute top-share difference is 22.5% and average difference is 8.9%.  |

|                            |                                                                                                                                                                                                                  |                                                                                                                                                                     |      |                                                                                                                       |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------------------------------------------------------------------------------------------------------|
| days_to_intervention_start | NULL / No Active Intervention: 60.0% Same day / 0 days: 3.0% 1–2 days: 5.0% 3–7 days: 10.0% 8–14 days: 10.0% 15–30 days: 8.0% 31–60 days: 3.0% >60 days: 1.0%                                                    | NULL / No Active Intervention: 0.0%; Same day / 0 days: 0.6%; 1–2 days: 0.6%; 3–7 days: 6.1%; 8–14 days: 8.6%; 15–30 days: 34.1%; 31–60 days: 50.0%; >60 days: 0.0% | Fail | Bucket distribution differs from expected; max absolute bucket difference is 60.0% and average difference is 18.3%.   |
| intervention_start_month   | same for this NULL / No Active Intervention: 60.0% January: 4.0% February: 3.2% March: 3.4% April: 3.2% May: 3.7% June: 3.2% July: 4.0% August: 3.4% September: 3.4% October: 3.2% November: 3.0% December: 2.8% | November: 9.9%; June: 9.8%; January: 9.0%; August: 8.7%; December: 8.5%; September: 8.3%; October: 8.3%; February: 8.1%                                             | Fail | Distribution shape differs from expected; max absolute top-share difference is 50.1% and average difference is 8.1%.  |
| intervention_start_wday    | NULL / No Active Intervention: 60.0% Monday: 8.0% Tuesday: 8.5% Wednesday: 8.5% Thursday: 8.0% Friday: 6.0% Saturday: 0.6% Sunday: 0.4%                                                                          | Monday: 25.0%; Tuesday: 22.0%; Wednesday: 21.0%; Thursday: 18.0%; Friday: 14.0%                                                                                     | Fail | Distribution shape differs from expected; max absolute top-share difference is 35.0% and average difference is 10.5%. |

## Dependencies

This dependency section validates the documented requirements directly. For requested probabilities, the decision is based on observed conditional treatment or outcome rates versus the requested probability. For dependency rules, the decision is based on conditional rates, outcome rates, odds ratios, relative risks, and mean lifts. Correlation coefficients are not used for pass/fail decisions because they do not validate whether the requested probabilities and dependency rules were implemented.

How to read this section: Fail means the documented dependency is missing, reversed, saturated, or materially miscalibrated; Weak means the direction is mostly right but the magnitude does not match

**Commented [RH12]:** 1. The treatment effect is largely increasing with risk and propensity. I don't see enough "high-risk/low-benefit" and "moderate-risk/high-benefit" members, which is central to PRISM. Nicky, I have a table named 'PRISM population segments' in the 'data generation specification' document. Could you make a similar table for the 1000-row data that GenRocket shared with us and compare the two? We would like more members with similar propensity but substantially different treatment effects

**Commented [ND13R12]:** I created the PRISM population segment table for the 1,000-row GenRocket dataset and compared it against the intended design. The generated data does not match the target distribution. High Risk + Low Benefit is 0% versus the 15% target, Moderate Risk + High Benefit is 9.6% versus the 20% target, and Low Risk + Low Benefit is 0% versus the 25% target. Instead, 59.2% of members fall into Low Risk + Moderate Benefit, which was not one of the intended PRISM segments. This supports your concern that treatment benefit is not sufficiently decoupled from risk/propensity, and that GenRocket should adjust the generation logic to better represent members with similar propensity but meaningfully different treatment effects

**Commented [ND14R12]:** I think we should ask GenRocket to clarify exactly how risk\_tier, Propensity\_Score, and true\_treatment\_effect were generated. This also ties back to the generation order: Baseline Variables → Propensity Score → Probabilistic Treatment Assignment. Treatment effect should be generated independently enough from risk/propensity so that members with similar propensity can still have substantially different expected benefit.

the specification; Pass means the direct conditional rates or mean lifts are consistent with the requested rule.]

**Commented [ND15]:** This section is now improved, no longer doing Spearman, but rather validated directly using conditional treatment rates, outcome rates, odds ratios, or relative risks.

## Dependency Issues

| Area                 | Check                                            | Result                                        | Expected                            | Status | Finding                                                                                                                                         |
|----------------------|--------------------------------------------------|-----------------------------------------------|-------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Treatment assignment | Overall intervention probability                 | 32.5%                                         | Approximately 40%                   | Fail   | Observed treatment assignment is compared directly with the requested overall probability.                                                      |
| Treatment assignment | Intervention probability for risk_tier=Low       | 18.5%                                         | 25.0% target                        | Fail   | Pass/fail is based on the conditional treatment rate for this tier.                                                                             |
| Treatment assignment | Intervention probability for risk_tier=Medium    | 49.1%                                         | 45.0% target                        | Pass   | Pass/fail is based on the conditional treatment rate for this tier.                                                                             |
| Treatment assignment | Intervention probability for risk_tier=High      | 83.0%                                         | 65.0% target                        | Fail   | Pass/fail is based on the conditional treatment rate for this tier.                                                                             |
| Treatment assignment | Intervention probability for risk_tier=Very High | 100.0%                                        | 65.0% target                        | Fail   | Pass/fail is based on the conditional treatment rate for this tier.                                                                             |
| Treatment assignment | ED visits last 6 months >=3 vs <3                | high=34.7%;<br>low=29.9%;<br>RR=1.16; OR=1.24 | Very high positive driver, weight 5 | Weak   | Using the clinically meaningful 3+ ED visit threshold, treatment lift is positive but much smaller than the documented weight-5 driver implies. |
| Treatment assignment | Admissions last 6 months >=1 vs 0                | high=34.9%;<br>low=27.1%;<br>RR=1.29; OR=1.44 | Very high positive driver, weight 5 | Weak   | Using the clinically meaningful any-admission threshold, treatment lift is                                                                      |

|                      |                                                                  |                                                 |                                                |      |                                                                                          |
|----------------------|------------------------------------------------------------------|-------------------------------------------------|------------------------------------------------|------|------------------------------------------------------------------------------------------|
|                      |                                                                  |                                                 |                                                |      | positive but much smaller than the documented weight-5 driver implies.                   |
| Treatment assignment | Clinical Score high vs low                                       | high=64.0%;<br>low=19.5%;<br>RR=3.29; OR=7.35   | High positive driver, weight 4                 | Pass | Conditional treatment rates validate the documented treatment driver.                    |
| Treatment assignment | Utilization Score high vs low                                    | high=46.3%;<br>low=19.9%;<br>RR=2.33; OR=3.49   | High positive driver, weight 4                 | Pass | Conditional treatment rates validate the documented treatment driver.                    |
| Treatment assignment | SDOH Score high vs low                                           | high=57.5%;<br>low=22.8%;<br>RR=2.52; OR=4.59   | High positive driver, weight 4                 | Pass | Conditional treatment rates validate the documented treatment driver.                    |
| Treatment assignment | Age high vs low                                                  | high=38.4%;<br>low=26.6%;<br>RR=1.44; OR=1.72   | Low positive driver; only slight lift expected | Weak | Age has a positive treatment lift, but the design calls for only a slight direct effect. |
| Treatment assignment | behavioral_health_risk_flag conditional intervention probability | yes=49.5%;<br>no=20.5%;<br>RR=2.42; OR=3.81     | High positive driver, weight 4                 | Pass | Decision uses direct treatment rates among members with and without the driver.          |
| Treatment assignment | chf_flag conditional intervention probability                    | yes=86.4%;<br>no=30.0%;<br>RR=2.88;<br>OR=14.76 | Moderate positive driver, weight 2             | Pass | Decision uses direct treatment rates among members with and without the driver.          |
| Treatment assignment | ckd_flag conditional intervention probability                    | yes=71.8%;<br>no=28.9%;<br>RR=2.49; OR=6.27     | Moderate positive driver, weight 2             | Pass | Decision uses direct treatment rates among members with and without the driver.          |

|                      |                                                        |                                             |                                     |      |                                                                                             |
|----------------------|--------------------------------------------------------|---------------------------------------------|-------------------------------------|------|---------------------------------------------------------------------------------------------|
| Treatment assignment | copd_flag conditional intervention probability         | yes=60.6%;<br>no=29.2%;<br>RR=2.07; OR=3.72 | Moderate positive driver, weight 2  | Pass | Decision uses direct treatment rates among members with and without the driver.             |
| Treatment assignment | diabetes_flag conditional intervention probability     | yes=55.0%;<br>no=26.8%;<br>RR=2.05; OR=3.33 | Moderate positive driver, weight 1  | Pass | Decision uses direct treatment rates among members with and without the driver.             |
| Treatment assignment | polypharmacy_flag conditional intervention probability | yes=67.0%;<br>no=25.1%;<br>RR=2.67; OR=6.06 | Moderate positive driver, weight 2  | Pass | Decision uses direct treatment rates among members with and without the driver.             |
| Treatment assignment | dual_eligible conditional intervention probability     | yes=40.4%;<br>no=29.9%;<br>RR=1.35; OR=1.59 | Moderate positive driver, weight 2  | Pass | Decision uses direct treatment rates among members with and without the driver.             |
| Treatment assignment | Gender direct treatment spread                         | min=29.4%;<br>max=66.7%;<br>spread=37.3%    | Minimal/no meaningful direct effect | Fail | Demographic or geographic fields should not materially drive treatment assignment directly. |
| Treatment assignment | Language direct treatment spread                       | min=28.3%;<br>max=45.2%;<br>spread=16.9%    | Minimal/no meaningful direct effect | Fail | Demographic or geographic fields should not materially drive treatment assignment directly. |
| Treatment assignment | Region direct treatment spread                         | min=32.5%;<br>max=32.5%;<br>spread=0.0%     | Minimal/no meaningful direct effect | Pass | Demographic or geographic fields should not materially drive treatment assignment directly. |
| Outcome generation   | Overall 90-day ED outcome probability                  | 9.8%                                        | Approximately 10%                   | Pass | Observed outcome prevalence is compared directly                                            |



|                    |                                               |                                                  |                                                                           |      |                                                                                                 |
|--------------------|-----------------------------------------------|--------------------------------------------------|---------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------------------|
|                    |                                               |                                                  |                                                                           |      | with the requested overall rate.                                                                |
| Outcome generation | 90-day ED probability for risk_tier=Low       | 0.0%                                             | 4.0% target                                                               | Fail | Pass/fail is based on the conditional outcome rate for this risk tier.                          |
| Outcome generation | 90-day ED probability for risk_tier=Medium    | 18.3%                                            | 10.0% target                                                              | Fail | Pass/fail is based on the conditional outcome rate for this risk tier.                          |
| Outcome generation | 90-day ED probability for risk_tier=High      | 51.1%                                            | 22.0% target                                                              | Fail | Pass/fail is based on the conditional outcome rate for this risk tier.                          |
| Outcome generation | 90-day ED probability for risk_tier=Very High | 66.7%                                            | 22.0% target                                                              | Fail | Pass/fail is based on the conditional outcome rate for this risk tier.                          |
| Outcome generation | ED outcome curve by prior ED visits           | 0=0.0%; 1=9.0%; 2=12.9%; 3+=12.0%                | Risk should rise gradually through 2 visits, then rise more rapidly at 3+ | Fail | The direct conditional outcome curve does not show the requested jump after 3+ prior ED visits. |
| Outcome generation | Prior ED visits >=3 vs <3                     | exposed=12.0%; reference=7.2%; RR=1.66; OR=1.75  | Relative risk about 4.00x                                                 | Fail | Outcome relationship is missing or reversed.                                                    |
| Outcome generation | Prior admissions >=1 vs 0                     | exposed=12.2%; reference=4.3%; RR=2.84; OR=3.10  | Relative risk about 3.00x                                                 | Pass | Observed conditional outcome rates are consistent with the requested relative risk.             |
| Outcome generation | Behavioral health risk yes vs no              | exposed=20.5%; reference=2.2%; RR=9.25; OR=11.39 | Relative risk about 1.60x                                                 | Fail | Outcome relationship is directionally positive but materially overstates the                    |

|                    |                                  |                                                            |                           |      |                                                                                                       |
|--------------------|----------------------------------|------------------------------------------------------------|---------------------------|------|-------------------------------------------------------------------------------------------------------|
|                    |                                  |                                                            |                           |      | requested relative risk.                                                                              |
| Outcome generation | Substance use disorder yes vs no | exposed=12.0%;<br>reference=8.9%;<br>RR=1.35; OR=1.39      | Relative risk about 1.50x | Pass | Observed conditional outcome rates are consistent with the requested relative risk.                   |
| Outcome generation | Housing instability yes vs no    | exposed=59.0%;<br>reference=4.3%;<br>RR=13.62;<br>OR=31.77 | Relative risk about 1.40x | Fail | Outcome relationship is directionally positive but materially overstates the requested relative risk. |
| Outcome generation | Transportation barrier yes vs no | exposed=54.1%;<br>reference=2.2%;<br>RR=24.32;<br>OR=51.82 | Relative risk about 1.25x | Fail | Outcome relationship is directionally positive but materially overstates the requested relative risk. |
| Outcome generation | Food insecurity yes vs no        | exposed=26.0%;<br>reference=5.8%;<br>RR=4.52; OR=5.76      | Relative risk about 1.20x | Fail | Outcome relationship is directionally positive but materially overstates the requested relative risk. |
| Outcome generation | CHF yes vs no                    | exposed=40.9%;<br>reference=8.4%;<br>RR=4.89; OR=7.58      | Relative risk about 1.40x | Fail | Outcome relationship is directionally positive but materially overstates the requested relative risk. |
| Outcome generation | CKD yes vs no                    | exposed=34.1%;<br>reference=7.5%;<br>RR=4.52; OR=6.35      | Relative risk about 1.35x | Fail | Outcome relationship is directionally positive but materially overstates the                          |

|                    |                                                          |                                                           |                              |      |                                                                                                       |
|--------------------|----------------------------------------------------------|-----------------------------------------------------------|------------------------------|------|-------------------------------------------------------------------------------------------------------|
|                    |                                                          |                                                           |                              |      | requested relative risk.                                                                              |
| Outcome generation | COPD yes vs no                                           | exposed=30.8%;<br>reference=7.4%;<br>RR=4.18; OR=5.59     | Relative risk<br>about 1.30x | Fail | Outcome relationship is directionally positive but materially overstates the requested relative risk. |
| Outcome generation | Diabetes yes vs no                                       | exposed=23.3%;<br>reference=6.4%;<br>RR=3.64; OR=4.44     | Relative risk<br>about 1.15x | Fail | Outcome relationship is directionally positive but materially overstates the requested relative risk. |
| Outcome generation | Poor medication adherence PDC<0.80 vs >=0.80             | exposed=37.8%;<br>reference=5.1%;<br>RR=7.36;<br>OR=11.21 | Relative risk<br>about 1.20x | Fail | Outcome relationship is directionally positive but materially overstates the requested relative risk. |
| Outcome generation | Polypharmacy yes vs no                                   | exposed=30.7%;<br>reference=5.3%;<br>RR=5.75; OR=7.85     | Relative risk<br>about 1.20x | Fail | Outcome relationship is directionally positive but materially overstates the requested relative risk. |
| Outcome generation | Age >65 vs <=65                                          | exposed=11.9%;<br>reference=8.9%;<br>RR=1.34; OR=1.39     | Relative risk<br>about 1.10x | Pass | Observed conditional outcome rates are consistent with the requested relative risk.                   |
| Treatment effect   | Transportation barrier should increase treatment benefit | with=0.34;<br>without=0.23;<br>lift=0.11                  | High positive benefit driver | Pass | Mean true treatment effect is compared directly between                                               |

|                  |                                                           |                                    |                                  |      |                                                                                              |
|------------------|-----------------------------------------------------------|------------------------------------|----------------------------------|------|----------------------------------------------------------------------------------------------|
|                  |                                                           |                                    |                                  |      | members with and without the driver.                                                         |
| Treatment effect | Housing instability should increase treatment benefit     | with=0.35; without=0.23; lift=0.12 | High positive benefit driver     | Pass | Mean true treatment effect is compared directly between members with and without the driver. |
| Treatment effect | Food insecurity should increase treatment benefit         | with=0.30; without=0.23; lift=0.08 | Moderate positive benefit driver | Pass | Mean true treatment effect is compared directly between members with and without the driver. |
| Treatment effect | Behavioral health risk should increase treatment benefit  | with=0.29; without=0.21; lift=0.08 | High positive benefit driver     | Pass | Mean true treatment effect is compared directly between members with and without the driver. |
| Treatment effect | Depression should increase treatment benefit              | with=0.32; without=0.23; lift=0.09 | Moderate positive benefit driver | Pass | Mean true treatment effect is compared directly between members with and without the driver. |
| Treatment effect | Anxiety should increase treatment benefit                 | with=0.32; without=0.23; lift=0.09 | Moderate positive benefit driver | Pass | Mean true treatment effect is compared directly between members with and without the driver. |
| Treatment effect | Medication nonadherence should increase treatment benefit | with=0.31; without=0.23; lift=0.08 | High positive benefit driver     | Pass | Mean true treatment effect is compared directly between members with and without the driver. |
| Treatment effect | Frequent ED utilization should increase treatment benefit | with=0.28; without=0.20; lift=0.09 | High positive benefit driver     | Pass | Mean true treatment effect is compared directly between members with and without the driver. |

|                               |                                                               |                                       |                                    |      |                                                                                                                                   |
|-------------------------------|---------------------------------------------------------------|---------------------------------------|------------------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------|
| Treatment effect              | Multiple chronic conditions should increase treatment benefit | with=0.29; without=0.24; lift=0.05    | Moderate positive benefit driver   | Pass | Mean true treatment effect is compared directly between members with and without the driver.                                      |
| Treatment effect              | No PCP utilization should increase treatment benefit          | with=0.21; without=0.25; lift=-0.04   | Moderate positive benefit driver   | Fail | Mean true treatment effect is compared directly between members with and without the driver.                                      |
| Treatment effect              | Pregnancy should have minimal treatment-benefit influence     | with=0.23; without=0.25; lift=-0.02   | Minimal/low benefit driver         | Pass | Observed treatment benefit is appropriately near-neutral for this low/minimal driver.                                             |
| Treatment effect              | Advanced CHF + CKD treatment benefit                          | advanced=0.30; others=0.24; lift=0.06 | Should have lower/negative benefit | Fail | Advanced CHF+CKD members show higher, not lower, true treatment benefit.                                                          |
| Treatment effect distribution | High Benefit share                                            | 21.4%                                 | 20.0% target                       | Pass | Benefit-group share is inferred from true_treatment_effect bands: high >=0.30, moderate 0.125-<0.30, low 0.025-<0.125, no <0.025. |
| Treatment effect distribution | Moderate Benefit share                                        | 73.8%                                 | 50.0% target                       | Fail | Benefit-group share is inferred from true_treatment_effect bands: high >=0.30, moderate 0.125-<0.30, low 0.025-<0.125, no <0.025. |
| Treatment effect distribution | Low Benefit share                                             | 4.8%                                  | 25.0% target                       | Fail | Benefit-group share is inferred from true_treatment_effect bands: high >=0.30, moderate 0.125-                                    |

|                               |                                  |                                      |                                          |      |                                                                                                                                           |
|-------------------------------|----------------------------------|--------------------------------------|------------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|
|                               |                                  |                                      |                                          |      | <0.30, low 0.025-<br><0.125, no <0.025.                                                                                                   |
| Treatment effect distribution | No Benefit share                 | 0.0%                                 | 5.0% target                              | Pass | Benefit-group share is inferred from true_treatment_effect bands: high >=0.30, moderate 0.125-<br><0.30, low 0.025-<br><0.125, no <0.025. |
| Treatment effect segments     | High Risk + High Benefit         | 8.1%                                 | 15.0% target                             | Fail | Segment share is validated directly from risk_tier and true_treatment_effect bands.                                                       |
| Treatment effect segments     | High Risk + Low Benefit          | 0.0%                                 | 15.0% target                             | Fail | Segment share is validated directly from risk_tier and true_treatment_effect bands.                                                       |
| Treatment effect segments     | Moderate Risk + High Benefit     | 9.6%                                 | 20.0% target                             | Fail | Segment share is validated directly from risk_tier and true_treatment_effect bands.                                                       |
| Treatment effect segments     | Moderate Risk + Moderate Benefit | 12.6%                                | 25.0% target                             | Fail | Segment share is validated directly from risk_tier and true_treatment_effect bands.                                                       |
| Treatment effect segments     | Low Risk + Low Benefit           | 4.6%                                 | 25.0% target                             | Fail | Segment share is validated directly from risk_tier and true_treatment_effect bands.                                                       |
| Clinical dependencies         | diabetes_flag -> ckd_flag        | yes=37.6%;<br>no=1.1%;<br>lift=36.5% | Diabetes should increase CKD probability | Pass | Decision uses direct conditional rates.                                                                                                   |

|                       |                                                  |                                       |                                                   |      |                                                               |
|-----------------------|--------------------------------------------------|---------------------------------------|---------------------------------------------------|------|---------------------------------------------------------------|
| Clinical dependencies | diabetes_flag -> polypharmacy_flag               | yes=49.5%;<br>no=9.5%;<br>lift=40.0%  | Diabetes should increase polypharmacy probability | Pass | Decision uses direct conditional rates.                       |
| Clinical dependencies | diabetes_flag -> specialist_visits_last_6m       | yes=2.33;<br>no=0.55; lift=1.78       | Diabetes should increase specialist visits        | Pass | Decision uses direct conditional means.                       |
| Clinical dependencies | diabetes_flag -> admits_last_6m                  | yes=2.00;<br>no=1.75; lift=0.25       | Diabetes should increase admissions               | Pass | Decision uses direct conditional means.                       |
| Clinical dependencies | chf_flag -> admits_last_6m                       | yes=2.25;<br>no=1.78; lift=0.47       | CHF should increase admissions                    | Pass | Decision uses direct conditional means.                       |
| Clinical dependencies | chf_flag -> ed_visits_last_6m                    | yes=3.18;<br>no=2.62; lift=0.57       | CHF should increase ED utilization                | Pass | Decision uses direct conditional means.                       |
| Clinical dependencies | chf_flag -> specialist_visits_last_6m            | yes=4.50;<br>no=0.74; lift=3.76       | CHF should increase specialist visits             | Pass | Decision uses direct conditional means.                       |
| Clinical dependencies | behavioral_health_risk_flag -> ed_visits_last_6m | yes=2.70;<br>no=2.60; lift=0.11       | BH risk should increase ED utilization            | Weak | Decision uses direct conditional means.                       |
| Clinical dependencies | behavioral_health_risk_flag -> admits_last_6m    | yes=1.85;<br>no=1.76; lift=0.09       | BH risk should increase admissions                | Weak | Decision uses direct conditional means.                       |
| Clinical dependencies | substance_use_flag -> ed_visits_last_6m          | yes=2.72;<br>no=2.61; lift=0.12       | SUD should increase ED utilization                | Weak | Decision uses direct conditional means.                       |
| Clinical dependencies | substance_use_flag -> admits_last_6m             | yes=1.87;<br>no=1.77; lift=0.11       | SUD should increase admissions                    | Weak | Decision uses direct conditional means.                       |
| Behavioral health     | behavioral_health_risk_flag -> med_adherence_pdc | yes=0.83;<br>no=0.92;<br>change=-0.09 | BH risk should reduce                             | Pass | Lower PDC among members with the driver validates medication- |

|                                |                                                         |                                       |                                                             |      |                                                                                       |
|--------------------------------|---------------------------------------------------------|---------------------------------------|-------------------------------------------------------------|------|---------------------------------------------------------------------------------------|
| dependencies                   |                                                         |                                       | medication adherence                                        |      | nonadherence dependency.                                                              |
| Behavioral health dependencies | substance_use_flag -> med_adherence_pdc                 | yes=0.85;<br>no=0.89;<br>change=-0.04 | SUD should reduce medication adherence                      | Pass | Lower PDC among members with the driver validates medication-nonadherence dependency. |
| Behavioral health dependencies | depression_flag -> med_adherence_pdc                    | yes=0.77;<br>no=0.90;<br>change=-0.13 | Depression should reduce medication adherence               | Pass | Lower PDC among members with the driver validates medication-nonadherence dependency. |
| Behavioral health dependencies | anxiety_flag -> med_adherence_pdc                       | yes=0.77;<br>no=0.90;<br>change=-0.13 | Anxiety should reduce medication adherence                  | Pass | Lower PDC among members with the driver validates medication-nonadherence dependency. |
| SDOH dependencies              | housing_instability_flag -> food_insecurity_flag        | yes=54.0%;<br>no=16.2%;<br>lift=37.8% | Housing instability should increase food insecurity         | Pass | Decision uses direct conditional rates.                                               |
| SDOH dependencies              | housing_instability_flag -> transportation_barrier_flag | yes=100.0%;<br>no=5.1%;<br>lift=94.9% | Housing instability should increase transportation barriers | Pass | Decision uses direct conditional rates.                                               |
| SDOH dependencies              | housing_instability_flag -> utilities_insecurity_flag   | yes=100.0%;<br>no=5.0%;<br>lift=95.0% | Housing instability should increase utilities insecurity    | Pass | Decision uses direct conditional rates.                                               |



|                          |                                                   |                                               |                                                                              |      |                                                                                                     |
|--------------------------|---------------------------------------------------|-----------------------------------------------|------------------------------------------------------------------------------|------|-----------------------------------------------------------------------------------------------------|
| SDOH dependencies        | transportation_barrier_flag -> pcp_visits_last_6m | yes=4.33; no=1.49; lift=2.83                  | Transportation barriers should reduce PCP utilization                        | Fail | Mean PCP utilization is higher for members with transportation barriers, so the direction is wrong. |
| SDOH dependencies        | transportation_barrier_flag -> ed_visits_last_6m  | yes=2.95; no=2.59; lift=0.37                  | Transportation barriers should increase ED utilization                       | Pass | Decision uses direct conditional means.                                                             |
| SDOH dependencies        | Multiple SDOH barriers -> ed_visits_last_6m       | 2+ barriers=2.91; no barriers=2.55; lift=0.36 | Multiple SDOH barriers should raise ED utilization                           | Pass | Decision compares members with 2+ SDOH barriers to members with no SDOH barriers.                   |
| SDOH dependencies        | Multiple SDOH barriers -> true_treatment_effect   | 2+ barriers=0.34; no barriers=0.22; lift=0.12 | Multiple SDOH barriers should raise treatment benefit                        | Pass | Decision compares members with 2+ SDOH barriers to members with no SDOH barriers.                   |
| Interaction dependencies | CHF + CKD -> ed_visits_last_6m                    | both=3.17; exactly one=2.62; lift=0.54        | CHF + CKD should produce substantially greater utilization than either alone | Pass | Decision compares members with both CHF and CKD to members with exactly one of the two conditions.  |
| Interaction dependencies | CHF + CKD -> outcome_ed_90d                       | both=38.1%; exactly one=33.3%; lift=4.8%      | CHF + CKD should produce greater ED risk than either alone                   | Pass | Decision compares members with both CHF and CKD to members with exactly one of the two conditions.  |
| Interaction dependencies | High ED + high SDOH -> true_treatment_effect      | both high=0.37; others=0.23; lift=0.14        | High ED utilization + high SDOH should produce larger                        | Pass | Decision uses direct outcome/benefit rates for the combined subgroup.                               |

|                          |                                                          |                                                    |                                                                            |      |                                                                                                         |
|--------------------------|----------------------------------------------------------|----------------------------------------------------|----------------------------------------------------------------------------|------|---------------------------------------------------------------------------------------------------------|
|                          |                                                          |                                                    | treatment benefit                                                          |      |                                                                                                         |
| Interaction dependencies | High ED + high SDOH -> outcome_ed_90d                    | both high=61.7%; others=5.2%; lift=56.5%           | High ED utilization + high SDOH should produce higher ED risk              | Pass | Decision uses direct outcome/benefit rates for the combined subgroup.                                   |
| Interaction dependencies | BH risk + poor adherence -> outcome_ed_90d               | BH+poor adherence=39.5%; BH only=12.4%; lift=27.1% | BH risk + medication nonadherence should increase ED risk                  | Pass | Decision compares the combined dependency subgroup to behavioral-health members without poor adherence. |
| Interaction dependencies | BH risk + poor adherence -> true_treatment_effect        | BH+poor adherence=0.32; BH only=0.27; lift=0.04    | BH risk + medication nonadherence should increase treatment responsiveness | Pass | Decision compares the combined dependency subgroup to behavioral-health members without poor adherence. |
| Burden dependencies      | Higher clinical/SDOH burden -> ed_visits_last_6m         | high burden=2.83; lower burden=2.56; lift=0.27     | Higher clinical and SDOH burden should increase utilization and cost       | Pass | Decision uses direct conditional means for higher-burden members.                                       |
| Burden dependencies      | Higher clinical/SDOH burden -> admits_last_6m            | high burden=1.94; lower burden=1.74; lift=0.20     | Higher clinical and SDOH burden should increase utilization and cost       | Pass | Decision uses direct conditional means for higher-burden members.                                       |
| Burden dependencies      | Higher clinical/SDOH burden -> observation_stays_last_6m | high burden=1.94; lower burden=1.74; lift=0.20     | Higher clinical and SDOH burden should increase                            | Pass | Decision uses direct conditional means for higher-burden members.                                       |

|                     |                                                   |                                                     |                                                                      |      |                                                                   |
|---------------------|---------------------------------------------------|-----------------------------------------------------|----------------------------------------------------------------------|------|-------------------------------------------------------------------|
|                     |                                                   |                                                     | utilization and cost                                                 |      |                                                                   |
| Burden dependencies | Higher clinical/SDOH burden -> total_cost_last_6m | high burden=326.63; lower burden=79.75; lift=246.88 | Higher clinical and SDOH burden should increase utilization and cost | Pass | Decision uses direct conditional means for higher-burden members. |

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#### How Many Total Rows Should we have?

I would recommend 20,000 row dataset because the ED outcome event rate is only 9.8%. With a 70/30 train/test split, 20,000 rows provides approximately 1,372 outcome events in training and 588 outcome events in testing. This gives about 24.5 events per predictor across the 56 pre-encoded predictors, which is above the commonly used 20 EPV threshold and provides a stronger basis for model training, dependency validation, and performance evaluation than the current 1,000-row dataset.